Gradient-Based DAG Structure Learning with Low-Rank Latent Confounding

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August 2026

Note: This document was prepared with AI assistance and has been reviewed and verified by the author.

Problem Formulation

Let

$$X \in \mathbb{R}^{N \times d}$$

contain N observations of d observed variables. We model the data using the latent-variable structural equation model

$$\boxed{X = XW + ZL^\top + E}$$

where

$$W \in \mathbb{R}^{d \times d}$$

is the weighted adjacency matrix among the observed variables,

$$Z \in \mathbb{R}^{N \times k}$$

contains realizations of k latent variables, and

$$L \in \mathbb{R}^{d \times k}$$

describes how those latent variables affect the observed variables. The latent contribution

$$C = ZL^\top \in \mathbb{R}^{N \times d}$$

is assumed to be low rank, with $\operatorname{rank}(C) \ll d$.

The model therefore separates the observed data into

$$\boxed{X = \underbrace{XW}_{\text{directed observed DAG}} + \underbrace{C}_{\text{latent confounding}} + \underbrace{E}_{\text{unexplained noise}}.}$$

The structure-learning problem can be written generally as

$$\boxed{\begin{aligned} \min_{W,C} \quad & \mathcal{L}(X - XW - C) + \lambda_W \|W\|_1 + \lambda_C \, \mathcal{R}_{\text{rank}}(C) \\ \text{s.t.} \quad & h(W) = 0, \\ & \operatorname{diag}(W) = 0. \end{aligned}} \tag{1}$$

Here, $\mathcal{L}(\cdot)$ is a general data-fit loss that need not assume Gaussian random variables, $\|W\|_1$ promotes a sparse directed graph, and $\mathcal{R}_{\text{rank}}(C)$ promotes a low-dimensional latent explanation. The NOTEARS acyclicity constraint is retained through

$$\boxed{h(W) = \operatorname{tr}\big(e^{W \circ W}\big) - d,}$$

so that $h(W) = 0$ if and only if W corresponds to a DAG.

A mathematically clean low-rank formulation uses the nuclear norm,

$$\mathcal{R}_{\text{rank}}(C) = \|C\|_*,$$

which gives

$$\boxed{\min_{W,C} \mathcal{L}(X - XW - C) + \lambda_W \|W\|_1 + \lambda_C \|C\|_* \quad \text{s.t. } h(W) = 0.} \tag{2}$$

This expresses the model directly as joint learning of a sparse DAG and a low-rank latent-confounding component.

Gradient-Friendly Factorized Form

For scalable gradient-based optimization, the low-rank component can be factorized directly as

$$C = ZL^\top.$$

The corresponding structure-learning problem is

$$\boxed{\begin{aligned} \min_{W,Z,L} \quad & \mathcal{L}(X - XW - ZL^\top) + \lambda_W \|W\|_1 \\ & \quad + \frac{\lambda_L}{2} \left(\|Z\|_F^2 + \|L\|_F^2 \right) \\ \text{s.t.} \quad & h(W) = 0. \end{aligned}} \tag{3}$$

The factorization is consistent with the nuclear-norm view through

$$\|C\|_* = \min_{C = ZL^\top} \frac{1}{2} \left(\|Z\|_F^2 + \|L\|_F^2 \right).$$

Thus, the factorized form provides a gradient-friendly implementation of low-rank regularization without requiring an explicit low-rank constraint or repeated singular-value decompositions.

General Data-Fit Loss

Define the residual

$$R(W, Z, L) = X - XW - ZL^\top.$$

The loss can be written in a distribution-agnostic form as

$$\boxed{\mathcal{L}(R) = \frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{d} \ell(R_{ij}),}$$

where the scalar loss ℓ is selected according to the assumed observation model. The methodological formulation therefore does not require Gaussian data.

For the initial Gaussian test case, choose

$$\ell(r) = \frac{1}{2} r^2,$$

which gives the least-squares objective

$$\boxed{\begin{aligned} \min_{W,Z,L} \quad & \frac{1}{2N} \|X - XW - ZL^\top\|_F^2 + \lambda_W \|W\|_1 \\ & + \frac{\lambda_L}{2} \left(\|Z\|_F^2 + \|L\|_F^2 \right) \\ \text{s.t.} \quad & h(W) = 0. \end{aligned}} \tag{4}$$

When $k = 0$, the latent term disappears and the formulation reduces to the linear NOTEARS problem,

$$\min_W \frac{1}{2N} \|X - XW\|_F^2 + \lambda_W \|W\|_1 \quad \text{s.t. } h(W) = 0.$$

This provides the desired no-latent-variable limiting case.

Interpretation of the Learned Structure

The matrix W represents the learned directed causal structure among the observed variables. A nonzero entry corresponds to a directed edge in the observed DAG. The loading matrix L represents the influence of latent variables on the observed variables. If latent variable z_r has nonzero loadings on both X_i and X_j, then the latent structure contains the common-cause pattern

$$X_i \leftarrow z_r \rightarrow X_j.$$

A useful observed-space summary of latent confounding is

$$\boxed{S_L = LL^\top.}$$

Large off-diagonal values of $(S_L)_{ij}$ indicate that variables X_i and X_j share latent-factor loading. The learned model can therefore be reported using both the directed observed graph W and a latent-confounding structure derived from $LL^\top$. After thresholding, these correspond conceptually to

$$X_i \to X_j \quad \Longleftrightarrow \quad W_{ij} \neq 0,$$

and

$$X_i \leftrightarrow X_j \quad \Longleftrightarrow \quad (LL^\top)_{ij} \neq 0.$$

This provides an ADMG-like observed representation while retaining the continuous factorized variables W, Z, and L during optimization.

Latent Dimension

Rather than optimizing directly over an unknown number of hidden nodes, choose a moderate upper bound $k_{\max}$ and use

$$Z \in \mathbb{R}^{N \times k_{\max}}, \qquad L \in \mathbb{R}^{d \times k_{\max}}.$$

Regularization then controls the effective latent dimension. Standard Frobenius regularization encourages small factors but does not necessarily remove entire latent dimensions. If explicit latent-dimension selection is desired, a group penalty can be added, for example

$$\lambda_g \sum_{r=1}^{k_{\max}} \|L_{:r}\|_2.$$

The effective latent dimension can then be defined by

$$\hat{k} = \#\left\{r : \|L_{:r}\|_2 > \epsilon\right\}.$$

This keeps the formulation compatible with gradient-based optimization while allowing unnecessary latent factors to be suppressed.

Augmented-Lagrangian Optimization

As in NOTEARS, the acyclicity constraint can be handled with an augmented Lagrangian. The unconstrained objective becomes

$$\boxed{\begin{aligned}\mathcal{J}(W, Z, L) = \mathcal{L}(X - XW - ZL^\top) + \lambda_W \|W\|_1 \\ + \frac{\lambda_L}{2}\left(\|Z\|_F^2 + \|L\|_F^2\right) \\ + \alpha h(W) + \frac{\rho}{2} h(W)^2.\end{aligned}}$$

The variables W, Z, and L can then be optimized using gradient-based methods, while the acyclicity multiplier is updated according to

$$\alpha \leftarrow \alpha + \rho h(W).$$

The resulting formulation preserves the main computational advantage of NOTEARS: causal structure learning is posed as a continuous differentiable optimization problem rather than a combinatorial search over DAGs, while the low-rank factorization captures latent confounding in the same optimization.

1 Identifiability of the Static Latent-Confounded Model

The static formulation in Equation (4) poses structure learning directly in terms of a sparse directed matrix W and a low-rank latent contribution $C = ZL^\top$. An immediate identifiability difficulty is that sparsity of W and low rank of C alone do not uniquely determine their allocation. For example, if $\Delta W = ae_i e_j^\top$ modifies a single directed edge, then

$$X\Delta W = aX_{:i}e_j^\top$$

has rank one. Consequently, part of a directed contribution can in principle be absorbed into the low-rank term without changing the data fit. Additional conditions are therefore required to distinguish sparse directed structure from latent shared structure.

The conditions developed below are motivated by the sparse-plus-low-rank identifiability literature. Chandrasekaran et al. showed that a sparse component and a low-rank component can be separated when their corresponding matrix varieties are sufficiently transverse, or incoherent [2]. In latent Gaussian graphical models, this idea yields an identifiable sparse conditional precision matrix together with a low-rank contribution induced by unobserved variables [1]. Frot et al. subsequently used this structure for causal discovery, showing that when the observed conditional graph is sparse and a small number of hidden variables have sufficiently widespread effects, the Markov equivalence class of the underlying observed DAG can be recovered consistently [3]. We now translate this argument to the present structural equation formulation.

1.1 Population Model and Assumptions

Consider one observation written in column-vector form,

$$x = W^{\star\top}x + L^\star z + \varepsilon, \tag{5}$$

where $W^\star \in \mathbb{R}^{d\times d}$ is the true weighted adjacency matrix among the observed variables and $z \in \mathbb{R}^k$ contains the unobserved common causes. Define

$$B^\star = I - W^{\star\top}.$$

Because $W^\star$ is acyclic, $B^\star$ is nonsingular after a topological permutation of the variables.

We impose the following assumptions.

A1. Acyclic linear latent-root SEM. The observed variables are generated according to Equation (5), where $W^\star$ is acyclic and

$$z \sim \mathcal{N}(0, \Sigma_z), \qquad \varepsilon \sim \mathcal{N}(0, D),$$

with $\Sigma_z \succ 0$, $D \succ 0$ diagonal, and $z \perp\!\!\!\perp \varepsilon$. The latent variables are exogenous common causes: interventions on the observed variables do not alter their distribution. This is the latent-root setting also considered by Frot et al. [3].

A2. Sparse conditional observed structure. The observed DAG has bounded local complexity. In particular, the conditional precision matrix

$$S^\star = (I - W^\star) D^{-1} (I - W^{\star\top}) \tag{6}$$

is sparse, with bounded or sufficiently slowly growing maximum degree. This condition is slightly stronger than sparsity of $W^\star$ itself because $S^\star$ reflects the moralized conditional graph associated with the linear SEM.

A3. Low-dimensional and pervasive latent influence. The number of latent variables satisfies $k \ll d$. Their effect on the observed variables is sufficiently diffuse that the resulting low-rank component does not itself resemble a sparse matrix. More precisely, if $H^\star$ denotes the low-rank matrix derived below and $H^\star = U \Lambda U^\top$, its singular subspace is assumed to satisfy an incoherence condition of the form

$$\max_{1 \le j \le d} \left\| U^\top e_j \right\|_2^2 \le \mu \frac{k}{d}, \tag{7}$$

for a moderate incoherence parameter μ. Thus, a hidden factor cannot concentrate its effect on only a small number of observed variables.

A4. Sparse–low-rank transversality. Let $\Omega(S^\star)$ denote the tangent space associated with the sparsity pattern of $S^\star$ and $\mathcal{T}(H^\star)$ the tangent space of the rank-k matrix variety at $H^\star$. We require

$$\Omega(S^\star) \cap \mathcal{T}(H^\star) = \{0\}, \tag{8}$$

with a quantitative separation such as

$$\left\| P_{\Omega(S^\star)} P_{\mathcal{T}(H^\star)} \right\|_{\mathrm{op}} \le 1 - \eta, \qquad \eta > 0. \tag{9}$$

Together with the standard Fisher-information regularity conditions, this is the rank–sparsity identifiability condition developed by Chandrasekaran et al. [2, 1]. Equivalent sufficient

conditions can be expressed through the product of sparsity and low-rank incoherence measures; Frot et al., for example, employ conditions of the form

$$\xi(H^\star)\mu(S^\star) \le c$$

for a constant determined by the information geometry of the model [3].

A5. Markov and faithfulness condition. Conditional on the latent variables, the distribution of x is faithful to the DAG $G(W^\star)$. Equivalently, $G(W^\star)$ is a perfect map of $p(x \mid z)$:

$$X_i \perp\!\!\!\perp X_j \mid X_A, z \quad \Longleftrightarrow \quad X_i \perp_{G(W^\star)} X_j \mid X_A$$

for every admissible conditioning set A. This is the same causal target used in the hidden-variable DAG framework of Frot et al. [3].

1.2 Theorem 1: Population Identifiability of the Observed DAG

Theorem 1. Suppose the data are generated according to Equation (5) and Assumptions A1–A5 hold. Then the marginal observational distribution $p(x)$ uniquely identifies the conditional independence model of $G(W^\star)$ and therefore identifies $G(W^\star)$ up to its Markov equivalence class:

$$p(x) \quad \Longrightarrow \quad [G(W^\star)]_{\mathrm{MEC}}. \tag{10}$$

Proof. From Equation (5),

$$B^\star x = L^\star z + \varepsilon,$$

and hence

$$x = B^{\star -1} L^\star z + B^{\star -1}\varepsilon.$$

Conditioning on z gives

$$\mathrm{Cov}(x \mid z) = B^{\star -1} D B^{\star -\top}.$$

Its inverse is therefore

$$\begin{aligned} K_{x|z} &= B^{\star\top} D^{-1} B^\star && (11)\\ &= (I - W^\star) D^{-1} (I - W^{\star\top}) && (12)\\ &= S^\star. && (13) \end{aligned}$$

Marginalizing the latent variables instead gives

$$\Sigma_x = B^{\star -1} \left(D + L^\star \Sigma_z L^{\star\top} \right) B^{\star -\top}. \tag{14}$$

Applying the Woodbury identity,

$$\left(D + L^\star \Sigma_z L^{\star\top} \right)^{-1} = D^{-1} - D^{-1} L^\star \left(\Sigma_z^{-1} + L^{\star\top} D^{-1} L^\star \right)^{-1} L^{\star\top} D^{-1}. \tag{15}$$

Consequently, the marginal precision matrix can be written as

$$K_x = \Sigma_x^{-1} = S^\star - H^\star, \tag{16}$$

where

$$H^\star = Q^\star \left(\Sigma_z^{-1} + L^{\star\top} D^{-1} L^\star\right)^{-1} Q^{\star\top}, \qquad Q^\star = (I - W^\star) D^{-1} L^\star. \tag{17}$$

The matrix $H^\star$ satisfies

$$H^\star \succeq 0, \qquad \operatorname{rank}(H^\star) \le k.$$

Thus the proposed latent-variable SEM induces the sparse-minus-low-rank precision decomposition

$$\boxed{K_x = \underbrace{S^\star}_{\text{sparse conditional precision}} - \underbrace{H^\star}_{\text{low-rank latent contribution}}.} \tag{18}$$

Under Assumptions A2–A4, the sparse and low-rank components are distinguishable from their sum. The rank–sparsity theory of Chandrasekaran et al. establishes precisely this type of identifiability: sparse conditional structure can be separated from low-rank hidden structure when the sparse and low-rank tangent spaces are sufficiently transverse and the relevant information-theoretic regularity conditions hold [2, 1]. Therefore the marginal distribution $p(x)$, which determines K_x in the Gaussian model, identifies $S^\star$.

It follows from Equation (13) that

$$S^{\star -1} = \operatorname{Cov}(x \mid z).$$

Hence $p(x)$ identifies the covariance matrix of the observed variables after conditioning on the latent common causes. For a nonsingular Gaussian distribution, the covariance matrix determines all conditional independence relations [4]. Assumption A5 then implies that these conditional independences coincide with the d-separation relations of $G(W^\star)$.

Finally, two DAGs encode the same set of conditional independence relations if and only if they have the same skeleton and the same unshielded collider structures [5]. They therefore belong to the same Markov equivalence class. Any alternative model satisfying A1–A5 and producing the same marginal distribution $p(x)$ must consequently have an observed DAG Markov equivalent to $G(W^\star)$. Thus

$$p(x) \Longrightarrow [G(W^\star)]_{\mathrm{MEC}},$$

which proves the claim. □

1.3 Relation to Existing Latent-Variable Identifiability Results

Theorem 1 connects the present structural formulation directly to the sparse-plus-low-rank theory of latent-variable graphical models. In the framework of Chandrasekaran et al., marginalization over a small number of latent Gaussian variables produces a precision matrix

that decomposes into a sparse conditional component and a low-rank positive-semidefinite correction [1]. Their identifiability conditions require that the conditional graphical model be sparse while the latent influence is spread across many observed variables. The geometric origin of these conditions is the rank–sparsity transversality principle developed in [2].

Frot et al. apply this principle explicitly to causal structure learning [3]. Their model is especially close to the assumptions above: conditional on the hidden variables, the observed variables follow a linear structural equation model with diagonal disturbance covariance; the hidden variables are exogenous; the observed causal DAG is sparse; and a small number of hidden variables act broadly across the observed system. They first estimate the sparse conditional precision matrix through a low-rank-plus-sparse decomposition and then apply a causally sufficient structure-learning algorithm to recover the corresponding CPDAG. Under their high-dimensional assumptions, the resulting estimator consistently recovers the Markov equivalence class of the observed DAG.

The target of Theorem 1 is deliberately similar. Although marginalizing hidden common causes would generally motivate an ancestral-graph representation, the present model retains a sparse DAG among the observed variables and treats the latent variables as exogenous common inputs. This choice is meaningful when the scientific target is the directed structure that would remain among the observed variables after accounting for the shared latent influence. Frot et al. make the same distinction and note that under pervasive hidden confounding a MAG or PAG can become dense and retain substantial orientation ambiguity, whereas the conditional observed DAG and its CPDAG can remain sparse and informative [3].

There is, however, an important difference between the existing sparse-plus-low-rank estimators and the method proposed here. Chandrasekaran et al. and Frot et al. perform the separation in covariance or precision space. The present method instead estimates the directed and latent components jointly in structural-equation space,

$$X = XW + C + E, \qquad C = ZL^\top,$$

while simultaneously enforcing acyclicity of W. Theorem 1 therefore establishes *population identifiability of the model class*; it does not by itself prove that the optimizer in Equation (4) converges statistically to that identifiable solution.

The remaining theoretical problem is consequently estimator consistency. A subsequent result must connect the direct decomposition $XW + C$ used by the proposed objective to the identifiable population model above. A natural starting condition is a transversality condition directly in structural space,

$$X\mathcal{T}_{W^\star} \cap \mathcal{T}_{C^\star} = \{0\}, \tag{19}$$

together with restricted curvature of the data-fit loss, minimum directed edge strength, minimum nonzero latent singular value, and suitable scaling of the sparsity and low-rank regularization parameters. Establishing this connection will form the estimator-consistency result developed separately.

1.4 Scope of the Identifiability Result

Theorem 1 concerns the Markov equivalence class of the directed structure among the observed variables after accounting for latent common causes. It does not require the individual latent factors themselves to be uniquely identified. In particular, the factorization $C = ZL^\top$ is invariant to invertible changes of latent coordinates, so the labels and coordinates of the latent variables are not population-identifiable without additional assumptions. This ambiguity does not affect Theorem 1 because its causal target is $[G(W^\star)]_{\mathrm{MEC}}$, while the latent contribution is used to separate shared hidden variation from the sparse observed directed structure.

Time-Series Extension

For time-series data, let

$$x_t \in \mathbb{R}^d, \qquad t = 1, \dots, T,$$

denote the observed variables at time t, and let p denote the maximum temporal lag included in the model. We introduce a contemporaneous interaction matrix $W_0 \in \mathbb{R}^{d\times d}$ and lagged interaction matrices $W_\tau \in \mathbb{R}^{d\times d}$ for $\tau = 1, \dots, p$. The latent confounding term is retained through time-varying latent realizations $z_t \in \mathbb{R}^k$ and a loading matrix $L \in \mathbb{R}^{d\times k}$.

The corresponding time-series structural equation model is

$$\boxed{x_t = W_0^\top x_t + \sum_{\tau=1}^{p} W_\tau^\top x_{t-\tau} + Lz_t + \epsilon_t.} \tag{20}$$

Here, W_0 represents contemporaneous directed relationships among the observed variables, while W_τ represents directed relationships from time $t-\tau$ to time t. The latent factor z_t captures unobserved influences at time t that can affect multiple observed variables through L.

To write the model over the full time series, define $X_0 \in \mathbb{R}^{(T-p)\times d}$ as the matrix of current observations and $X_\tau \in \mathbb{R}^{(T-p)\times d}$ as the corresponding observations shifted by lag τ. Let $Z \in \mathbb{R}^{(T-p)\times k}$ collect the latent realizations. Equation (20) then becomes

$$\boxed{X_0 = X_0 W_0 + \sum_{\tau=1}^{p} X_\tau W_\tau + ZL^\top + E.} \tag{21}$$

The low-rank contribution $ZL^\top$ therefore plays the same role as in the independent-realization formulation, while the matrices $W_1, \dots, W_p$ account explicitly for temporal dependence.

Define the time-series residual

$$R_{\mathrm{TS}} = X_0 - X_0 W_0 - \sum_{\tau=1}^{p} X_\tau W_\tau - Z L^\top.$$

The gradient-friendly structure-learning problem is

$$\begin{aligned} \min_{W_0,\ldots,W_p,Z,L} \quad & \mathcal{L}(R_{\mathrm{TS}}) + \lambda_0 \|W_0\|_1 + \lambda_{\mathrm{lag}} \sum_{\tau=1}^{p} \|W_\tau\|_1 \\ & + \frac{\lambda_L}{2} \left(\|Z\|_F^2 + \|L\|_F^2 \right) \\ \text{s.t.} \quad & h(W_0) = 0, \\ & \operatorname{diag}(W_0) = 0. \end{aligned} \tag{22}$$

Only the contemporaneous matrix W_0 requires the NOTEARS acyclicity constraint. Lagged edges point from past samples to future samples and therefore cannot create a directed cycle within a single time slice. This permits recurrent relationships across time while retaining an acyclic contemporaneous structure.

As in the independent-realization formulation, $\mathcal{L}(\cdot)$ can be any suitable differentiable data-fit loss. For the initial Gaussian test case, the least-squares specialization is

$$\begin{aligned} \min_{W_0,\ldots,W_p,Z,L} \quad & \frac{1}{2(T-p)} \left\| X_0 - X_0 W_0 - \sum_{\tau=1}^{p} X_\tau W_\tau - Z L^\top \right\|_F^2 \\ & + \lambda_0 \|W_0\|_1 + \lambda_{\mathrm{lag}} \sum_{\tau=1}^{p} \|W_\tau\|_1 \\ & + \frac{\lambda_L}{2} \left(\|Z\|_F^2 + \|L\|_F^2 \right) \\ \text{s.t.} \quad & h(W_0) = 0. \end{aligned} \tag{23}$$

When $k = 0$, the latent-confounding term disappears and Equation (22) reduces to a NOTEARS-style dynamic structure-learning problem with contemporaneous and lagged directed interactions. Thus, the time-series formulation preserves the same low-rank latent-confounding extension while replacing independent realizations with an explicitly autoregressive structural model.